studied ships' compasses, then he made a model globe out of lodestone (a magnetic rock) and used it to test compass needles. He concluded that Earth is a giant magnet and that its north and south magnetic poles more or less correspond with its geographic ones. Gilbert was Queen Elizabeth I's physician for the last 2 years of her reign. He died soon after the queen, probably from the plague.

GIORDANO BRUNO

1548–1600

Burned alive as a heretic, Giordano Bruno believed that the Universe had no center and that there are many worlds, and many suns, just like ours. The Italian philosopher, mathematician, and one-time monk made the Sun seem insignificant with his discovery that it is just one star surrounded by planets and moons, among an infinite number of others. In doing so, he rejected the traditional Earth-centered cosmology, went further than the Sun-centered one described by Nicolaus Copernicus, and came up with a more accurate picture of an infinite Universe.

JOHN NAPIER

1550–1617

Born near Edinburgh, Scotland, John Napier joined St. Andrew's University in 1563, but left a year later to continue his mathematical studies in Europe. On his return, he wrote an interpretation of the Bible's Book of Revelation and designed various war weapons. But Napier's best-known achievement is his invention of the logarithm, a number that represents another number in order to simplify a difficult division or multiplication sum. He also created "Napier's bones," a set of rods that could be used to multiply or divide numbers.

FRANCIS BACON

1561–1626

After studying law, the Englishman Francis Bacon became a member of parliament in 1581. He was unpopular with Queen Elizabeth I, but found favor with James I and rose to become Lord Chancellor in 1618. Two years later, he was convicted of bribery and banned from public office. Undeterred, he pursued his interest in science and published *Novum Organum* (*New Instrument*) in 1620. Denouncing Aristotle's approach as unscientific, Bacon outlined his scientific method: make an observation, develop a theory that might explain that observation, and test it using experiments. It has been the mainstay of science ever since.

WILLEBRORD SNELL

1580–1626

Astronomer and mathematician Willebrord Snell was born in Leiden, the Netherlands, and succeeded his father as professor of mathematics at the university there. Although he wrote on astronomy, navigation, and geodesy (ways of measuring the Earth), he is best known for discovering the law of refraction in 1621. When light passes through air, it travels in a straight line; but when it passes through denser materials, it changes direction—that is, it is refracted. Snell's law relates how much the light bends to the properties of the material it is passing through.

PIERRE GASSENDI

1592–1655

Philosopher, mathematician, and priest Pierre Gassendi criticized his fellow Frenchman René Descartes' description of a mechanical Universe filled with particles of matter and proposed an alternative view that was more compatible with his Christian beliefs. He suggested that atoms share some of the properties of the physical things they make up, such as shape and size, and that atoms may join together to make larger molecules. Unlike Descartes, he believed in the existence of vacuums, going as far as to say that most matter consisted of "void."

EVANGELISTA TORRICELLI

1608–1647

After studying in Rome, Italian physicist and mathematician Evangelista Torricelli worked as an assistant to Galileo Galilei for the last 3 months of the astronomer's life. At Galileo's suggestion, Torricelli filled a tube with mercury and upended it into a bowl of mercury. When some of the mercury stayed in the tube, he realized that the space above it was a vacuum and that the height of the mercury in the tube was determined by atmospheric pressure. He had created the first barometer. A unit of pressure, the torr, is named after him.

MARIA CUNITZ

1610–1664

Growing up in Schweidnitz, Silesia (now Swidnica, Poland), Maria Cunitz had no formal education but was taught by her father, a doctor. Her second husband, Elias von Löwen, also a doctor, taught her mathematics and medicine and encouraged her interest in astronomy. The couple fled Silesia during the Thirty Years' War, and Cunitz spent her time in exile working on a set of astronomical tables that simplified Johannes Kepler's *Rudolphine Tables* (1627). She published her work in the book *Urania Propitia*. A fire destroyed Cunitz's home in 1656, and most of her papers were lost.